

DP Computer Science: A 2.1.5 TCP/IP Model (HL Only)

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- Explain the **purpose** and **importance** of the TCP/IP model in network communication
- Identify and describe the functions of each layer in the TCP/IP model (**Application, Transport, Internet, Network Interface**)
- Explain the concepts of **encapsulation** and **decapsulation** in the context of data transmission
- Analyse real-world scenarios and relate them to the functions of the TCP/IP model

Key Vocabulary

Term	Definition
TCP/IP Model	A four-layer model describing how data is transmitted over networks
Encapsulation	The process of adding headers (and sometimes trailers) to data as it moves down the layers
Decapsulation	The process of removing headers as data moves up the layers
Header	Information added to the front of a data packet at each layer (source/destination addresses, protocol info, etc.)
Payload	The actual data being transmitted (the "contents" of the packet)
Packet	A unit of data transmitted over a network
Protocol	A set of rules governing how data is transmitted and received

IB ATL Skills

Skill	Description
Communication	Communicating information and ideas effectively using appropriate terminology
Collaboration	Working collaboratively with peers to complete activities
Thinking	Applying thinking skills critically and creatively to analyse the TCP/IP model
Information Literacy	Finding and interpreting information about network communication

2. The TCP/IP Model Overview

What is the TCP/IP Model?

The TCP/IP model is the foundation of Internet communication, dividing the process into four layers. It provides a standard framework for how data should be formatted, transmitted, routed, and received across networks.

The Four Layers



3. TCP/IP Model Layers in Detail

Layer 1: Application Layer

Aspect	Description
Function	Provides network services to user applications
Role	The interface between the user application and the network
What it does	Formats data for the application; establishes communication with remote applications
Key Protocols	HTTP, HTTPS, FTP, SMTP, DHCP, DNS
Data Unit	Message / Data
Examples	Web browsing (HTTP), email (SMTP), file transfer (FTP)

Analogy: The Application Layer is like the **front desk** of a hotel—it handles requests from guests and directs them to the right services.

Layer 2: Transport Layer

Aspect	Description
Function	Ensures reliable data transfer between applications on different devices
Role	Manages end-to-end communication; provides error checking and flow control
What it does	Breaks data into segments; adds port numbers; chooses TCP or UDP based on requirements
Key Protocols	TCP (reliable, connection-oriented), UDP (fast, connectionless)
Data Unit	Segment (TCP) / Datagram (UDP)
Examples	Web browsing uses TCP; streaming uses UDP

Analogy: The Transport Layer is like the **shipping department**—it chooses the best delivery method (TCP = registered mail, UDP = standard post) and ensures packages are labelled correctly.

Layer 3: Internet Layer

Aspect	Description
Function	Routes data between different networks
Role	Provides logical addressing (IP addresses) and routing
What it does	Adds source and destination IP addresses; determines the best path for data
Key Protocol	IP (Internet Protocol)
Data Unit	Packet
Examples	Data is routed from a home network to a server across the internet

Analogy: The Internet Layer is like the **postal sorting system**—it reads addresses and decides which route packages should take.

Layer 4: Network Interface Layer

Aspect	Description
Function	Handles the physical transmission of data over the network medium
Role	Converts data into signals for transmission; manages hardware addressing
What it does	Adds MAC addresses; transmits data as electrical signals, radio waves, or light
Key Protocols	Ethernet, Wi-Fi, PPP

Aspect	Description
Data Unit	Frame
Examples	Data is sent over a Wi-Fi signal or through an Ethernet cable

Analogy: The Network Interface Layer is like the **delivery truck**—it physically carries packages to the destination address.

4. Encapsulation and Decapsulation

Encapsulation: Data Moving Down the Layers

Encapsulation is the process of adding headers (and sometimes trailers) to data as it moves down the TCP/IP layers.



Decapsulation: Data Moving Up the Layers

Decapsulation is the process of removing headers as data moves up the TCP/IP layers.



The Data Transformation at Each Layer

Layer	Input	Process	Output
Application	User data	Creates data for transmission	Data (payload)
Transport	Data	Adds port numbers; chooses TCP/UDP	Segment/Datagram
Internet	Segment/Datagram	Adds source/destination IP addresses	Packet
Network Interface	Packet	Adds MAC addresses; converts to signals	Frame

5. Protocols and Their Layers

Protocol	Full Name	Layer	Function
HTTP	Hypertext Transfer Protocol	Application	Web page transfer (unencrypted)
HTTPS	HTTP Secure	Application	Secure web page transfer
FTP	File Transfer Protocol	Application	File transfer
SMTP	Simple Mail Transfer Protocol	Application	Sending email
DNS	Domain Name System	Application	Domain name resolution
DHCP	Dynamic Host Configuration Protocol	Application	Automatic IP assignment
TCP	Transmission Control Protocol	Transport	Reliable data delivery
UDP	User Datagram Protocol	Transport	Fast, lightweight data delivery
IP	Internet Protocol	Internet	Logical addressing and routing
Ethernet	—	Network Interface	Wired network transmission
Wi-Fi	—	Network Interface	Wireless network transmission

6. TCP/IP vs. OSI Model

OSI Model (7 Layers)	TCP/IP Model (4 Layers)
Application	↕ Application
Presentation	
Session	
Transport	↕ Transport
Network	↕ Internet
Data Link	↕ Network Interface
Physical	

Key Difference: The TCP/IP model is simpler and more practical, combining the OSI model's top three layers (Application, Presentation, Session) into a single Application Layer.

7. Real-World Scenario: Sending an Email

Step-by-Step Through the Layers

Step	Layer	Action
1	Application	Email client (Gmail, Outlook) creates the email message; SMTP protocol is used
2	Transport	TCP is used to ensure reliable delivery; port numbers are added (port 25 for SMTP)
3	Internet	Source and destination IP addresses are added; data is routed across the internet
4	Network Interface	MAC addresses are added; data is converted to electrical/radio signals and transmitted
5	Physical	Signals travel across network cables/wirelessly to the destination

Encapsulation in Action

text

SENDING AN EMAIL: ENCAPSULATION

Application:

[Email Content]

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Transport:

[TCP Header][Email Content] → Segment

• Source Port: 5678

• Destination Port: 25

• Sequence Number: 1 of 3

↓

Internet:

[IP Header][TCP Header][Email Content]

• Source IP: 192.168.1.10

• Destination IP: 172.217.160.142

↓

Network Interface:

[MAC][IP][TCP][Email Content][MAC Trailer]

• Source MAC: 00:1A:2B:3C:4D:5E

• Destination MAC: 60:71:82:93:A4:B5

↓

Physical:

Data is transmitted as signals

8. Lesson Sequence

Lesson Plan

Phase	Time	Activity
Engage	5 min	"Broken Telephone" analogy—miscommunication happens; need for clear rules and protocols

Phase	Time	Activity
Explore	15 min	TCP/IP model overview; "Layer Roles" brainstorm—groups identify key functions for each layer
Explain	20 min	Layer deep dive; "Data Packet Relay Race" — simulate encapsulation and decapsulation
Elaborate	15 min	Real-world scenarios (sending email, streaming video); identify protocols used at each layer
Wrap-up	5 min	Key takeaways; exit ticket

Activity: "Layer Roles" Brainstorm

Instructions:

1. Divide students into four groups
2. Assign each group one layer of the TCP/IP model
3. Each group brainstorms:
 - o The key function of their layer
 - o Protocols used at their layer
 - o What type of data unit is handled
 - o A real-world analogy
4. Groups share their findings with the class

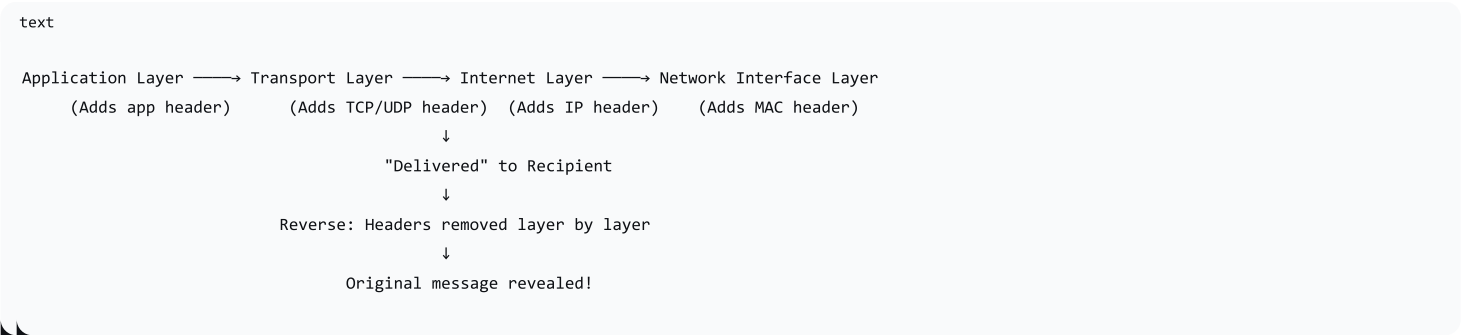
Activity: "Data Packet Relay Race"

Purpose: Simulate encapsulation and decapsulation.

Setup:

- Divide students into four teams (Application, Transport, Internet, Network Interface)
- Prepare a "data packet" (small object or paper with a message)
- Provide blank headers for each team

The Race:



Example Headers Students Can Add:

Layer	Header Example
Application	"Email Client, SMTP, Text"
Transport	"TCP, Source Port 5678, Dest Port 25, Packet 1 of 3"
Internet	"Source IP 192.168.1.10, Dest IP 172.217.160.142"

Layer	Header Example
Network Interface	"Source MAC 00:1A:2B:3C:4D:5E, Dest MAC 60:71:82:93:A4:B5"

Debrief Questions:

1. What information did each layer add to the packet?
2. Why is it important that data is encapsulated at each layer?
3. What would happen if the headers were not removed in the correct order?

9. Assessment Activities

Assessment Type	Description
Class Participation	Observe engagement and contributions during discussions
"Layer Roles" Brainstorm	Assess understanding of layer functions
Data Packet Relay Race	Assess understanding of encapsulation and decapsulation
TCP/IP in Action Analysis	Apply knowledge to real-world scenarios
Exit Ticket	Gauge understanding and identify areas needing clarification
Multiple Choice Quiz	eBook page A2.1.5 quiz

10. Differentiation

Level	Strategy
Support	Provide visual aids, concise definitions, and real-world examples
Challenge	Encourage advanced students to research more complex networking concepts (subnetting, IPv6, network security protocols)

11. Resources

- Presentation (Slide Deck)
- eBook (A2.1.5 with quiz)
- Quizlet (A2.1.5 vocabulary flashcards)

12. Summary: Key Takeaways

Concept	Key Point
Purpose	The TCP/IP model provides a standard framework for network communication
Four Layers	Application, Transport, Internet, Network Interface
Application	Provides services to applications (HTTP, FTP, SMTP, DHCP)

Concept	Key Point
Transport	Ensures data transfer (TCP = reliable, UDP = fast)
Internet	Routes data between networks (IP addressing)
Network Interface	Handles physical transmission (Ethernet, Wi-Fi)
Encapsulation	Adding headers as data moves down the layers
Decapsulation	Removing headers as data moves up the layers

| "The TCP/IP model is the foundation of the Internet—it ensures that data can be transmitted reliably, efficiently, and securely across different networks."